

Nick Buraglio

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Experience

Co-Chair IETF IPv6 Operations working group - 2023 - Present

- Co-chair the [operational IPv6 working group \(v6ops\)](#) for the Internet Engineering Task Force (IETF). Solicit input from network operators and users to identify operational issues with IPv6 networks, and determine solutions or workarounds to those issues. Solicit discussion and documentation of the issues and opportunities in IPv6-only operation, and of the resulting innovations. Full Charter can be found [here](#)

Planning and Architecture Group, Energy Sciences Network (ESnet) - 2013 - Present

- Member of the planning and architecture team responsible for strategy and architecture for an international high performance network. Responsibilities include:
 - Provide prototyping, support, assessment, and expertise for new and disruptive technologies
 - Act as security strategist for internal and externally facing services
 - Guide framework and policy on change management process
 - Present to community as a subject-matter expert on various topics including but not limited to IPv6, performance analytics, Segment Routing, and open perimeter security architectures
 - Manage vendor relationships
 - Mentor students and junior engineers
 - Lead teams of engineers with diverse disciplines in order to accomplish large, high visibility projects
 - Lead Department of Energy agency level IPv6-only implementation team lead for [OMB M-21-07](#)

Consulting Research Architect, Office of the CIO, University of Illinois - 2025 - Present

- Provide expertise on IPv6, cloud strategy, and WAN architecture for the University of Illinois System on a contractual basis.

Lead Network Engineer, University of Illinois, ICCN, UC2B, PeoriaNEXT - 2008 - 2013

- Team lead and network architect for regional optical network connecting all University of Illinois campuses, NCSA, and administrative networks to each other and external resources. Responsibilities included
 - Backup optical engineer for the InterCampus Communication Network (ICCN)
 - Team lead, strategist and architect for campus research networking initiatives including Science DMZ architecture and deployment
 - Primary engineer, architect, and representative for PeoriaNEXT, a layer2/3 internet exchange for Peoria, IL region

- Team lead and primary network architect for UC2B, a BTOP funded fiber to the premises ISP and NATOA broadband project of the year awardee
- Provide direction, technical expertise and mentorship to junior network engineers
- Guest lecturer for computer science department in IT networking and security courses

SCinet - 2003, 2005-2006, 2009, 2011-2012, 2014-2016, 2018-2019, 2023

- Participation in SCinet working group in various roles including wireless, routing, network security, SDN, and UNIX services
- Team lead for SC15 SDN team
- Develop, deploy and maintain OpenFlow based, production SDN architecture for SC15 network
- Develop, deploy, and coordinate 100G SDN QoS scheduling production testbed for SC16 network
- Team Lead for [production heterogeneous SDN network](#) (2018)
- Group Lead for Experimental Networks Group - XNet (2019)
- Subject-matter expert for [IPv6-mostly](#) initiative (2023)

Network Engineer, National Center for Supercomputing Applications - 2002 - 2008, 2012 - 2013

- Member of a small team responsible for all Layer1 -- Layer3 services
 - Network engineering liaison and advisor to incident responders and network security team, including management of high capacity Bro IDS systems
 - Collaborated with FBI on NCDIR project to assist and train field agents in network security and black hat techniques
 - Managed network based UNIX services including DNS, DHCP, RADIUS and NetFlow collectors
 - Design and maintain network monitoring infrastructure

Senior Network Engineer, IT Architect, Sol Tec Internet services - 2000 - 2002

- Manager for all technical teams consisting of full and part time employees including system engineers, network engineers, and help desk professionals
 - Provided technical leadership and strategy to fast growing regional internet service and early broadband provider
 - Authored and executed a detailed architecture expansion plan that included tripling capacity and relocation of all transit, infrastructure, UNIX, and colocation services
 - Designed, deployed, and supported extensive broadband services throughout greater central Illinois and southern Chicago areas
 - Managed all vendor relationships

Freelance IT Architect and Technical Writer - 1998 - Present

- Provide consultation, design, and strategic input for data center and service provider networks, including:
 - Metro, middle-mile, and fiber to the X (FTTx), design, configuration, maintenance
 - Transit and interconnect
 - IX design, implementation, and support
 - Extensive support for BGP, DNS, Automation (Ansible, Puppet), network analytics (Netflow, sFlow, long term baseline creation), capacity planning, performance tuning.
 - Backbone architecture (PNI, IX peerings, Traffic Engineering, MPLS, Segment Routing),

- Security architecture
- IPv6 planning and integration
- Freelance technical writing

Additional work history and certifications available upon request

Education, Certifications, Awards, Contributions, Publications

- B.S. Illinois State University
- Co-Author of [RFC 9637](#)
- Co-Author of [RFC 9872](#)
- Co-host of [IPv6 Buzz podcast](#), part of the Packet Pushers Podcast Network.
- Recipient of Letter of Commendation from U.S. Department of Energy Office of Science
- Department of Justice top secret clearance (2008)
- Co-Principal Investigator for NSF CC-NIE grant award [#1341025](#)
- Co-author, collaborator and sub-awardee for NSF CICI award [1642142](#)
- Principal Investigator (PI) for ESnet / Berkeley Lab collaboration with Office of Veterans Affairs project [MVP Champion Project](#)
- [IPSpace AG](#) Technical advisory board member, webinar host, subject matter expert
- [RouteViews Project](#), Member of the Board of Advisors
- Member of the technical advisory board for the [CAIDA Global Measurement Infrastructure to Improve Internet Security project](#)
- [NSF](#) Panel Reviewer
- [NTIA](#) Grant reviewer
- Co-author of IETF Drafts on subjects ranging from IPv6 to Segment Routing. [Full list](#).
- Co-Authored publication [CoreFlow: Enriching Bro security events using network traffic monitoring data](#)
- Co-Authored ACM Publication [Machine learning-based analysis of COVID-19 pandemic impact on US research networks](#)
- Co-Authored publication [Toward live inter-domain network services on the ExoGENI testbed](#)
- Received U.S. patent application serial no. 18/052,614 TITLE: AUTONOMOUS TRAFFIC (SELF-DRIVING) NETWORK WITH TRAFFIC CLASSES AND PASSIVE/ACTIVE LEARNING (Nov-2022)
- Member of the program committee for [the TNC Conference Series for 2023-2024](#)
- Consulting Information Technology Architect; Advisory Board Member [Albuquerque Internet Exchange](#) (ABQIX)
- Experienced public speaker, technical instructor, and international collaborator
- [Additional Publications](#)